Contamination Model: Results, Timing and Comparison

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Summary

On simulated data, the model caught every contaminated batch (120/120) about **7.8 h after contamination started**. That is **about 7.5 h earlier** than the smell or colour check MaverickX relies on today. Within a further 4–8 h it names contamination as the leading explanation.

On data we did not write, the results held up as follows:

- **IndPenSim** (simulated industrial penicillin batches): 1 false alarm in 30 normal batches, 8 of 10 equipment faults caught (median 5 h after the fault started), and no fault mistaken for contamination.
- **Real erythromycin plant data** (406 batches): batches the model flagged did turn out worse, with significantly lower final potency.

Caveat. The synthetic numbers are a software demonstration, not proven real-world performance. None of the external datasets contains contamination, so real contamination detection still needs MaverickX's batch records.

What the model does and how each dataset was used

The model learns what a normal batch looks like hour by hour, from approved normal batches. It then follows each new batch and acts in four steps:

1. **Detect:** raise an alert when a signal moves outside normal batch-to-batch variation.
2. **Diagnose:** rank the possible explanations, which are contamination, slow growth, a measurement fault or a process change.
3. **Recommend:** suggest the next check that best tells those explanations apart.
4. **Trace:** look for shared materials or equipment that link contaminated batches.

It never gives a contamination probability, and it never names an organism from process signals alone. Only a lab test confirms contamination.

Dataset	What it is	Learned from	Tested on	What it tests
Synthetic α-amylase	Our own simulation: normal, slow-growth, contaminated, sensor-spike and probe-drift batches	30 normal batches per dataset	25 later batches × 20 fresh datasets (500 batches), none used for tuning	Whether the whole system works as designed, including contamination and source tracing
IndPenSim	Published simulation of a 100,000 L industrial penicillin process, with labelled equipment faults	60 normal batches	30 normal + 10 fault batches	False alarms on realistic variation, how fast real deviations are caught, and whether non-biological faults are mistaken for contamination
Erythromycin plant	406 real industrial batches from 2022, no labels	Earliest 243 batches	Later 163 batches	Whether flagged batches really turn out worse (final potency was held back and used only to check)

Results per dataset

Each dataset answers a different question, so the stats differ. Results on synthetic data are a software demonstration. They are not real-world detection performance.

Synthetic α-amylase (seeds 11–30, 500 test batches)

Stat	Result	What it means
Contaminated batches flagged	120 / 120	Every contamination was caught (sensitivity 100%).
Sensor and probe faults flagged	120 / 120	Measurement problems are caught too, so they can be told apart rather than missed.

Stat	Result	What it means
Normal and slow-growth batches flagged	81 / 260	Often a brief alert on legitimate variation; slow growth is unusual by design.
False alerts per test batch	1.9	Mostly short, isolated alerts that a scientist can dismiss quickly.
Contamination named as leading explanation	12 / 120 at first alert, 82 at +4 h, 114 at +8 h, 120 at end of batch	The model waits for enough evidence before naming a cause. Early on it says "unresolved" and recommends a discriminating check.
Wrong leading explanation for contamination	5 at first alert, 2 at +8 h, 0 at end of batch	How often the report would point the investigation the wrong way.
Planted contaminated medium lot found as top lead	20 / 20 datasets	Source tracing finds a shared contaminated material.
False strong source leads	1 / 20 datasets	A genuine chance cluster, which is why tracing reports leads, not proof.

Why 120/120 is not proof. We wrote both the simulator and the rule book, so they share the same assumptions about how contamination looks. The result shows the software works as designed, not that it would catch real contaminants.

IndPenSim (40 held-out batches)

Stat	Result	What it means
Normal batches flagged	1 / 30	Low false-alarm rate across three control styles. The one alert was read as a likely measurement glitch.
Faults caught	8 / 10	Real process deviations are detected.
Median fault onset → first alert	5 h	Detection speed on realistic process data.
Alerts before any fault started	0	No premature alarms.

Stat	Result	What it means
Batches wrongly called contamination	0 / 40	Equipment faults are not mislabelled as contamination.
Base-flow faults caught	0 / 2	The pH controller hides this fault, so we need to monitor controller outputs.

This dataset also exposed two flaws, both now fixed. The first was scoring hours where too few reference batches existed. It had caused two false contamination calls, and fixing it cut the median detection time from 20.5 h to 5 h. The second was a one-hour clock offset. Because both fixes were found on this test set, the IndPenSim numbers are no longer fully unseen.

Erythromycin plant (163 later batches, real data)

Stat	Result	What it means
Logger errors flagged at ingestion	2,016 readings, none deleted	Bad sensor values are marked, not silently removed.
Batches flagged for review	14 / 163 (9%)	A manageable review load.
Flagged among low-potency batches	8 / 48 (17%) vs 6 / 115 (5%) among the rest	Flags concentrate on worse batches.
Median final potency: flagged vs not	10,434 vs 10,848	Flagged batches really turned out worse.
Chance of this by luck	Fisher p = 0.02, Mann-Whitney p = 0.001	The link is unlikely to be chance, though only 14 batches were flagged.
Called contamination	0 / 14	Diagnoses stayed cautious: 13 of 14 led with "process-condition change".

The limits came out very wide on this data, so only large deviations alert. Our planned fix is to set limits from a high percentile instead of the single worst reference batch.

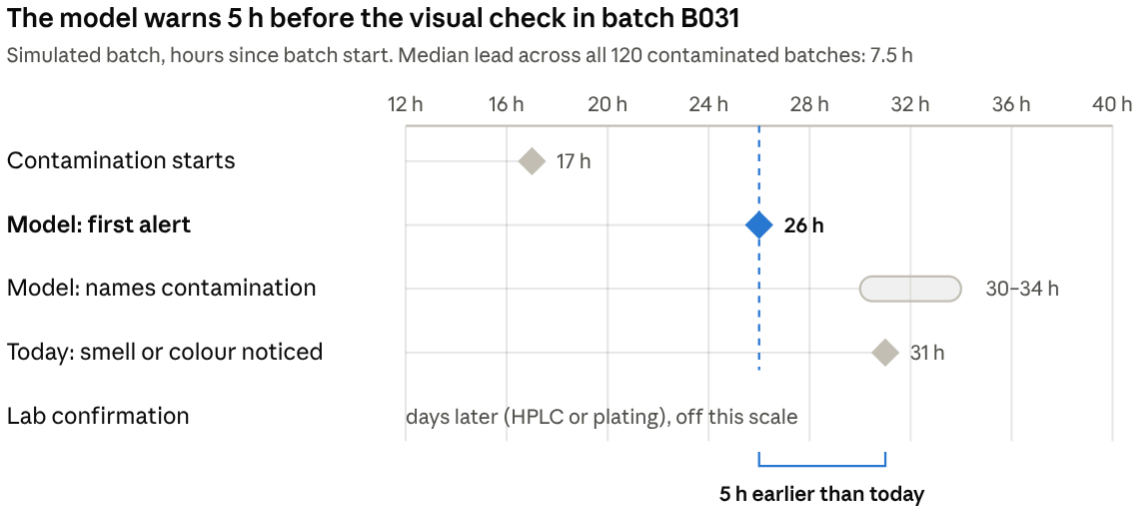
Time taken and time saved

The model warns about **7.5 h before today's visual check** on simulated data. It names contamination about 4–8 h after that first warning.

Timing measure	What it shows	Synthetic	IndPenSim
Detection delay (problem starts → first alert)	**Speed**: how sensitive the detectors are	~7.8 h median	5 h median
Lead time (model alert vs today's method)	**Effectiveness**: time saved compared with current practice	~7.5 h earlier (5–9 h across datasets)	No comparison available
Alert → confident diagnosis	**Usefulness**: when the warning says what is likely wrong	+4 h for 82/120, +8 h for 114/120	No contamination to name

Timing is not accuracy on its own. A model that alarmed every hour would have zero delay and be useless. Timing must be read with the detection rate, the false-alarm rate and diagnosis correctness. Lowering the limits gives earlier alerts but more false alarms. Ours are tuned towards fewer false alarms.

Ideal vs real vs model time



timeline of one simulated contaminated batch (B031)

In this batch the model alerted 9 h after contamination started and 5 h before the visual check. A confident diagnosis came at about the same time as the visual check.

- **Real onset:** when contamination actually starts (14–24 h in the simulation). It is known only in simulation, never on real plant data.
- **Ideal detection:** the earliest hour the deviation rises above normal variation. This is not hour 0: the contaminant starts tiny (about 0.02 against a maximum of about 3) and grows exponentially. Sampling also limits it, since sensors are hourly and lab assays

come back hours late. What matters most is warning before the point where the batch can no longer be saved.

- **Model detection:** about 7.8 h after onset, then about 4–8 h more to a confident diagnosis.

Why some problems take longer (IndPenSim)

Fault	Hours to alert	Why
Aeration	0	Changes dissolved oxygen immediately
Substrate feed	5–6	The effect builds up through metabolism
Coolant flow	10–11	The temperature controller partly compensates
Base flow	never caught	The pH controller hides it completely

Contamination behaves like the slower cases: its effect builds up gradually, and controllers can mask early signs. That is why we plan to monitor controller outputs too.

Comparison with the current method

Today MaverickX notices contamination by smell, colour and experience, often after the batch is already lost. The model adds an earlier, consistent, explained warning. It does not replace lab confirmation.

	Current method	With the model
How contamination is noticed	Smell, colour, operator experience	Continuous comparison of every signal against normal batches
When it is noticed	~12–18 h after onset in our simulation; often too late in practice	~7.8 h after onset (simulated), ~7.5 h earlier
What you learn	"Something is wrong"	Which signals changed, by how much, ranked explanations and the evidence for each
What to do next	Experience, or an expensive lab test	A recommended check chosen to tell the explanations apart
Confirmation	HPLC or plating: days; HPLC about $10,000 per test or $500 per 10 samples	Still needs the same lab test; the model only targets it earlier and better

	Current method	With the model
Finding the source	Manual review	Automatic check of shared medium lots, inoculum lots and bioreactors
Consistency	Depends on who is on shift	Same rules every time, versioned and auditable
Missed or false calls	Not measured	Measured: 120/120 caught and 1.9 false alerts per batch (synthetic); 1 false alarm in 30 batches (IndPenSim)

Limits and what we need from MaverickX

The model has not yet been tested on real contamination. That needs MaverickX's batch records.

- The synthetic results share our own assumptions about how contamination looks, so they are a demonstration, not evidence of real performance.
- The IndPenSim faults are equipment faults, not contamination. Two flaws were fixed after testing on that dataset.
- The rule book is a student draft that MaverickX scientists have not yet reviewed.
- The limits are too wide on real plant data. The planned fixes are percentile-based limits, monitoring controller outputs, and an "equipment fault" explanation.

Questions for MaverickX:

- [ ] At what hour does contamination typically start (after inoculation, a feed or sampling)?
- [ ] At what hour do operators usually notice it today?
- [ ] After what hour is a batch unsalvageable, and what can still be done before then?
- [ ] How long do confirmatory tests take in practice, and what do they cost?
- [ ] Can we have batch records that include confirmed contamination cases?

With these answers we can state the result that matters: how many hours before the point of no return the model warns, and at what false-alarm rate.